

1) Project Title

Design and Development of a Full-Stack AI-Enabled Worker Safety and Hazard Detection System

2) Statement About the Problem

Industrial workplaces such as construction sites, factories, warehouses, and mines involve various safety risks. Accidents often occur due to unsafe working conditions, lack of real-time monitoring, delayed hazard identification, and human negligence. Traditional safety measures rely heavily on manual supervision, which may not always detect hazards in time.

Many workplaces lack intelligent systems that can continuously monitor worker environments and identify potential safety violations or hazardous situations. As a result, preventable accidents lead to injuries, loss of life, production delays, and financial losses. There is a strong need for a smart, automated system that can assist in improving worker safety through early hazard detection.

3) Why Is This Particular Topic Chosen

- Worker safety is a **major concern in industrial and construction sectors**
- Increasing adoption of **AI-based monitoring systems**
- Strong **social responsibility and ethical relevance**
- Government focus on **workplace safety regulations**
- Combines **AI + Full-Stack development**
- Practical and impactful for a **BCA final-year project**

4) Objective and Scope

Objectives

- To develop an AI-based system for detecting workplace hazards
- To identify unsafe conditions and safety violations using AI
- To provide real-time alerts and safety insights
- To help organizations improve compliance with safety standards
- To reduce workplace accidents through early detection

Scope

- Focus on industrial and construction work environments
- Detect common hazards such as unsafe zones, missing safety gear, or risky behavior
- Web-based dashboard for monitoring and alerts
- Suitable for supervisors, safety officers, and management
- Prototype-level implementation for academic purposes

5) Methodology

1. **Requirement Analysis**
 - Study workplace safety challenges and accident scenarios
2. **System Design**
 - Design frontend, backend, AI module, and database architecture
3. **Data Collection**
 - Use sample datasets or simulated images/videos representing workplace scenarios
4. **AI Hazard Detection**
 - Apply computer vision techniques to identify unsafe conditions
 - Classify hazards using trained machine learning models
5. **Full-Stack Development**
 - Develop dashboards to display alerts, hazard logs, and reports
6. **Testing & Validation**
 - Validate detection accuracy and system performance

6) Hardware & Software Requirements

Hardware Requirements

- Laptop / Desktop Computer

- Webcam or sample video input
- Minimum 4GB RAM
- Internet connection

Software Requirements

- Frontend: HTML, CSS, JavaScript (React optional)
- Backend: Python (Django / Flask)
- Database: MySQL / SQLite
- AI/ML: Python, OpenCV, Scikit-learn
- Development Tools: VS Code
- Operating System: Windows / Linux

7) Contributions of the Project to Society

- Improves **worker safety and accident prevention**
- Supports safer working environments
- Assists organizations in safety monitoring
- Reduces human dependency for hazard supervision
- Promotes responsible use of AI in industrial safety
- Contributes to overall employee well-being

8) Schedule of the Project

Phase	Activity	Duration
Phase 1	Problem study & requirement analysis	1 Week
Phase 2	System design & planning	1 Week
Phase 3	Data collection & AI model design	2 Weeks
Phase 4	Full-stack development	3 Weeks

Phase 5 Integration & testing 2 Weeks

Phase 6 Documentation & presentation 1 Week

Project Phase	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Requirement Analysis & Safety Study	✓	✓								
System Architecture & Database Design		✓	✓							
Dataset Collection & Preprocessing			✓	✓						
AI Hazard Detection Model Development				✓	✓	✓				
Frontend Dashboard Development					✓	✓				
Backend & Alert Management System						✓	✓			
System Integration & Testing							✓	✓		
Performance Evaluation & Optimization								✓	✓	
Documentation & Final Presentation									✓	✓

9) Limitations of the Project

- Detection accuracy depends on dataset quality
- Real-time processing may be limited by hardware
- Complex hazards may not be fully detectable
- Prototype-level implementation
- Environmental conditions may affect AI performance

10) References

1. Occupational Safety and Health Administration (OSHA) Guidelines
2. OpenCV Documentation
3. Python Machine Learning – Scikit-learn
4. AI in Workplace Safety – IEEE Research Papers
5. Django Official Documentation